

ROI analysis

in Nilearn

Some guidelines

https://visualneuroscience.github.io/dicom_to_glm/roi_analysis.html

Note that the code provided there is rather complex and usually the number of ROIs, contrasts and conditions will be much smaller

Generate contrast estimate maps from the GLM

```
# compute and write out contrast maps that will be used in the ROI analysis
# Note: we are using effect_size maps instead of z-stat

contrast_string_coh = "coh_audio"
# compute the contrast
con_coh = models[0].compute_contrast(contrast_string_coh, output_type="effect_size")
con_coh.to_filename('contrasts/sub-001_contrast-coherent_statmap-effectsizes.nii.gz')

contrast_string_inc = "inc_audio"
# compute the contrast
con_inc = models[0].compute_contrast(contrast_string_inc, output_type="effect_size")
con_inc.to_filename('contrasts/sub-001_contrast-incoherent_statmap-effectsizes.nii.gz')
```

Subject loop

```
from nilearn.maskers import NiftiLabelsMasker
results_rois = []

for sub in subs:
    labels_file = f"/home/jovyan/2026_ANI/lobule9_bids_correct/derivatives/fastsurfer/sub-{sub}"
```

```

masker=NiftiLabelsMasker(labels_img=labels_file, labels=label_names, standardize=False)
for contrast in contrast_names:
    print(f"Extracting contrast: {contrast} for subject: {sub}")
    label_values = masker.fit_transform(f"/home/jovyan/2026_ANI/glm/contrasts/sub-{sub}")

    # Convert masker values to long format
    for label_idx, value in enumerate(label_values): # Assuming masker_values is 2D with
        results_rois.append({
            "subject_id": sub,
            "contrast": contrast,
            "label_idx": label_idx,
            "label_name": label_names[label_idx],
            "value": value
        })

```

Save as csv

```

import os
import pandas as pd
# Save results_main as a CSV file
output_dir = "/home/jovyan/2026_ANI/glm"
os.makedirs(output_dir, exist_ok=True)
df_main = pd.DataFrame(results_rois)
<!-- df_main.to_csv(f"{output_dir}/results_roi.csv", index=False) -->
print(f"Results saved to {output_dir}/results_roi.csv")

```

Terminal 1 X lobule9_nilearn_ffx_volume.ipynb ROI_analysis.ipynb results_roi.csv X					
Delimiter: ., v					
	subject_id	contrast	label_id	label_name	value
1	001	coherent	0	-Cerebellum-White-Matter	-0.1943073
2	001	coherent	1	Left-Cerebellum-Cortex	-0.17599243
3	001	coherent	2	-Cerebellum-White-Matter	-0.14184229
4	001	coherent	3	Right-Cerebellum-Cortex	-0.14436291
5	001	coherent	4	Cbm_Left_I_IV	-0.16963518
6	001	coherent	5	Cbm_Right_I_IV	-0.17819777
7	001	coherent	6	Cbm_Left_V	-0.2373206
8	001	coherent	7	Cbm_Right_V	-0.2623769
9	001	coherent	8	Cbm_Left_VI	-0.20305718
10	001	coherent	9	Cbm_Vermis_VI	-0.2545001
11	001	coherent	10	Cbm_Right_VI	-0.22629727
12	001	coherent	11	Cbm_Left_CrusI	-0.20060171
13	001	coherent	12	Cbm_Right_CrusI	-0.19494818
14	001	coherent	13	Cbm_Left_CrusII	-0.19304459
15	001	coherent	14	Cbm_Right_CrusII	-0.20764457
16	001	coherent	15	Cbm_Left_VIIb	-0.16376883
17	001	coherent	16	Cbm_Right_VIIb	-0.16240378
18	001	coherent	17	Cbm_Left_VIIIa	-0.24222487
19	001	coherent	18	Cbm_Right_VIIIa	-0.21175109
20	001	coherent	19	Cbm_Left_VIIIb	-0.291845
21	001	coherent	20	Cbm_Right_VIIIb	-0.25551888
22	001	coherent	21	Cbm_Left_IX	-0.3044485
23	001	coherent	22	Cbm_Vermis_IX	-0.24312682
24	001	coherent	23	Cbm_Right_IX	-0.2739685
25	001	coherent	24	Cbm_Left_X	-0.43484443
26	001	coherent	25	Cbm_Vermis_X	-0.2652649
27	001	coherent	26	Cbm_Right_X	-0.15504715
28	001	incoherent	0	-Cerebellum-White-Matter	-0.20675515
29	001	incoherent	1	Left-Cerebellum-Cortex	-0.21124819
30	001	incoherent	2	-Cerebellum-White-Matter	-0.18301152
31	001	incoherent	3	Right-Cerebellum-Cortex	-0.18227077
32	001	incoherent	4	Cbm_Left_I_IV	-0.22472693

Plot

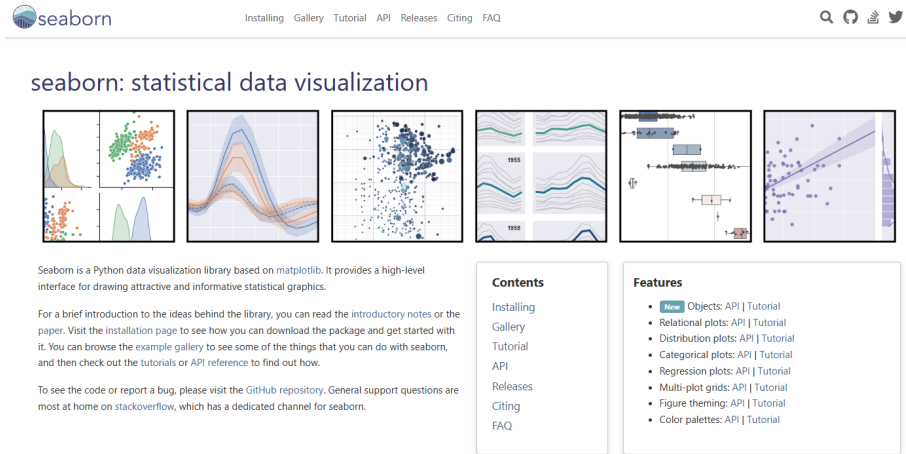
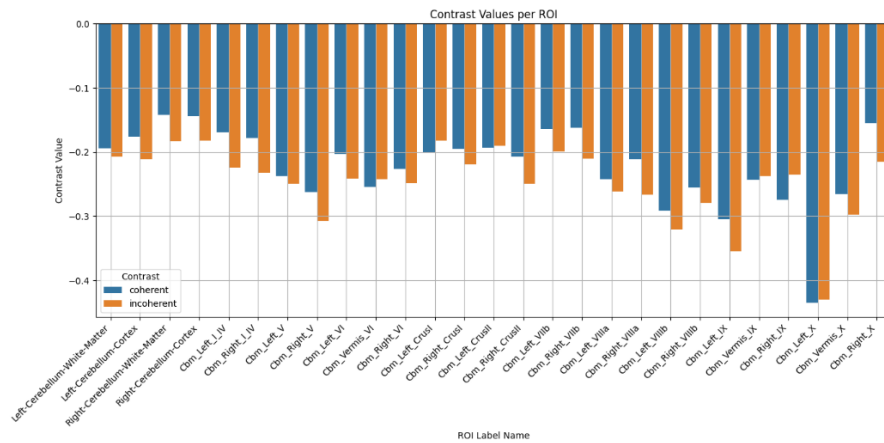


Figure 1: <https://seaborn.pydata.org/>

Result



References